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EDUCATION

- **Harvard University** Boston, MA
M.S. in Health Data Science August 2025 – March 2027
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 August 2021 – May 2025
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, R
- **ML systems:** PyTorch, TensorFlow, scanpy, AnnData, YAML
- **Cluster and automation:** SLURM, FASRC Cannon, job arrays, checkpoint/resume
- **Verification:** unittest, pytest
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

ENGINEERING PROJECTS

- **Closed-Loop Cluster Autoresearch: scGen vs CellOT** Boston, MA
Sole author — Harvard FASRC Cannon June 2026 – Present
 - Built a self-directed loop (submit → watch → triage → reflect → decide → continue → synthesize) that ran **338** SLURM jobs (318 ok, 20 handled errors) over **79.3** hours on Cannon.
 - Split a deterministic one-delta planner, gated by a validation policy, from a checkpointed research agenda. Optional LLM planner infrastructure includes token and cost accounting and a spend ceiling that falls back to the rule planner.
 - Routed tiny-MLP training to CPU partitions because GPU carries 200–500× fairshare cost, and resumed preempted jobs from the last checkpoint.
- **Human-Supervised Browser Automation System** Boston, MA
Sole author July 2026 – Present
 - Built a human-supervised browser automation system for job applications that keeps GitHub as durable policy and evidence, a Google Sheet as live runtime state, and Simplify as the application ledger.
 - Compiles YAML policy into one runtime file the local browser operator reads. Submit is allowed only after listed identity and writing checks.
 - Browser steps follow a compiled action sheet rather than rediscovering the form. Designed recovery for Mac shutdown, Wi-Fi loss, and browser restart.

RESEARCH EXPERIENCE

- **speciesOT: Cross-Species Single-Cell Translation** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute February 2026 – Present
 - Built a hub CLI that catalogs trained runs, ranks them on decoded-space gap-closed rather than raw R^2 , and prints SLURM chains without submitting jobs. Fifteen YAML specs are the intent source of truth.
 - Engineered atlas-scale I/O so the documented **43 GB** Tabula files are not fully loaded on a login node, and baked gene-axis plus scGen-shift sidecars so a new mouse prediction does not reopen the training atlas.
 - On a paired held-out-species evaluation, scGen R^2 was **0.85–0.90** with the source species in training versus **0.65–0.67** when that species was held out. Gene-space ranking reversed with split (6619 orthologs vs 1000 HVGs).